2627-MA140: Week 01, Lecture 2 (L02)

Draft

more Exers.

More About Functions

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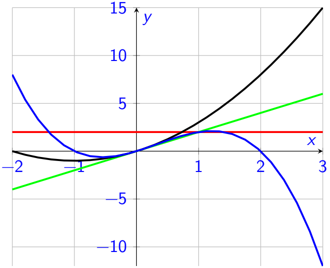

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The function of today's class is to learn about:

For more, see Sections 1.1 and 1.2 of https://math.libretexts.org/Bookshelves/Calculus/Calculus_(OpenStax)/01%3A_Functions_and_Graphs

Functions: notation

$f(x) = x^2$

Recall: This section is all about **functions**, which a "rule" for mapping inputs to outputs.

1. Writing $f : A \to B$ means the inputs come from the set A, and the outputs come from the set B. (A **set** is just a collection of things).
2. A is called the **domain**, and B is called the **co-domain**.
3. $y = f(x)$ means "x gets mapped to y according to the rule defined by f". We sometimes also say "y is the image of x".
4. The subset of B that contains all the images of the things in A is called the **range** of f.
5. When we write $x \in A$ we mean "x is an element of X, or "x belongs to A".

$f(x) = x^2$ with A as the set of (real) numbers and B is also the set of reals.

The Real Numbers, $\mathbb{R}$

In almost every example we'll consider, in MA140, f will a mapping form one **real** number to another.

The set of **real** numbers, denoted $\mathbb{R}$, are the numbers use to represent continuous quantities, such as length, speed, time and temperature.

- **continuous** means that there are no gaps, or "jumps": as a car accelerates from, say, 0m/s to 20m/s, at some point it will have taken on every speed between these values; where it has had .
- real numbers can be positive or negative.

Often, the domain of a function is not expilicitly stated.
In such a case the following **Domain Convention** applies.

The **domain** of a function f is the set of all numbers x for which $f(x)$ *makes sense* and gives a *real-number output*.

Example

1. Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \dfrac{1}{x^2 - x}$.

Note: $x=2$, $f(2) = \dfrac{1}{4-2} = \dfrac{1}{2}$ ✓

$x=1$ $f(1) = \dfrac{1}{1-1} = \dfrac{1}{0}$ ✗ not defined

so $x=1$ is not in the domain of f

because it leads to division by zero.

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Example

1. Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \dfrac{1}{x^2 - x}$.

x is in the domain of f_1 so long as $f(x)$ "makes sense": in this case no division by zero.

$$f_1(x) = \frac{1}{x(x-1)}$$

$x(x-1) = 0$ if $x = 0$ or $x - 1 = 0$.

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Example

1. Find the subset of $\mathbb{R}$ that is the **domain** of $f_1(x) = \dfrac{1}{x^2 - x}$.

So the domain of f_1 is all of $\mathbb{R}$ except 0 and 1.
That is: $\mathbb{R} / \{0, 1\}$.

Example

Find the subset of $\mathbb{R}$ that is the **domain** of the function $f_2(x) = \sqrt{x+2}$.

Since we work with the real numbers, $\mathbb{R}$, we can't have the square root of a negative number.

Eg $f_2(2) = \sqrt{2+2} = \sqrt{4} = 2$ ✓

$f_2(0) = \sqrt{0+2} = \sqrt{2}$

$f_2(-4) = \sqrt{-4+2} = \sqrt{-2}$ ✗.

So x is in the domain of f if $x+2 \geq 0$

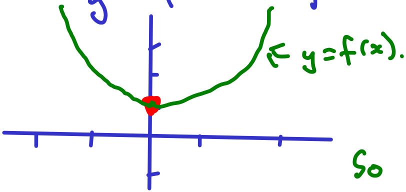
So $x \geq -2$. Same as $[-2, \infty)$

Example

Given the function $f_3(x) = 3x^2 + 1$, find the largest subset of $\mathbb{R}$ that is the domain of f_3. What is the corresponding **range**?

① Since $f(x) = 3x^2+1$ is defined for all $x \in \mathbb{R}$, then the domain is (all of) $\mathbb{R}$

② Range: set of values y for which $y = f(x)$ for some x.



$f(0) = 1$

$f(1) = 4$

$f(-1) = 4.$

So range is $[1, \infty)$ ie $1 \leq y < \infty$

Example

Identify the domain (in $\mathbb{R}$) and range of
$f_4(x) = \sqrt{(x+4)(3-x)}$

For x to be in the domain of f we need

$(x+4)(3-x) \geqslant 0$

So we need either (i) $x+4 \geq 0$ and $3-x \geq 0$

or (ii) $x+4 \leq 0$ and $3-x \leq 0$.

(i) $x+4 \geq 0 \Rightarrow x \geq -4$. $3-x \geq 0 \Rightarrow 3 \geq x$

$\Rightarrow x \leq 3$.

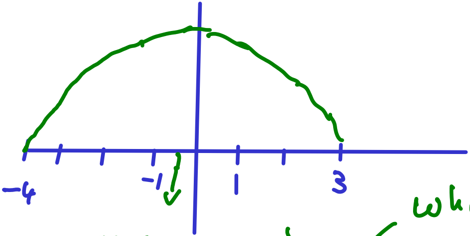
So any x with $-4 \leq x \leq 3$ is in the domain.

(ii) $x \leq -4$ and $x \geq 3$. Not possible!

Example

Identify the domain (in $\mathbb{R}$) and range of
$f_4(x) = \sqrt{(x+4)(3-x)}$

For the range we need to find all values of $f_4(x)$ for $-4 \leq x \leq 3$.



max at $x = -\frac{1}{2}$. why?

(come back in 2 weeks!).

Example

Identify the domain and range of $f_5(x) = \dfrac{1}{x}$.

Domain is all of $\mathbb{R}$ except $x = 0$.

So the domain is $\underbrace{(-\infty, 0) \cup (0, \infty)}_{\text{set notation!}}$ ie $\mathbb{R} / \{0\}$.

Range:

$y = f(x)$:

so the range too is $\mathbb{R} / \{0\}$.

4 Ways to Represent a Function

A function can be represented in different ways:

1. **verbally** (by a description in *words*);
2. **numerically** (as a *table* of values);
3. **visually** (as a *graph*);
4. **algebraically** (by an explicit *formula*).

Often it is possible, and useful, to go from one way to another.

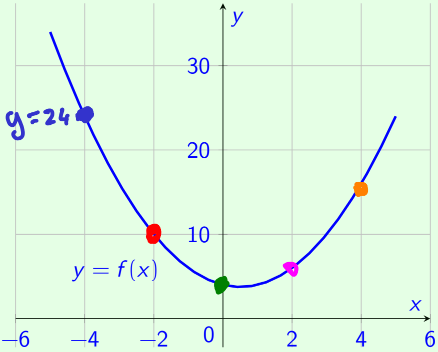
Especially from algebraic to visual.

Graphical Representation

Graph $\to$ Table

A common way to *visualize* a function $f : X \to \mathbb{R}$ is its *graph* in the x, y-plane. In this example, $f(x) = x^2 - x + 4$.

$f(-4) = 16 - (-4) + 4 = 24$



x	$f(x)$
-4	24
-2	10
0	4
2	6
4	16

A Catalog of Functions

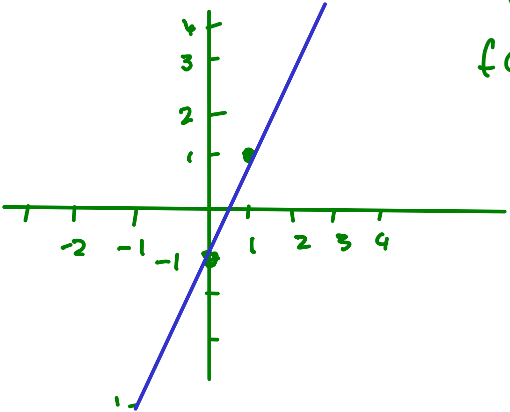
There are many *different types of functions* that can be used to *model relationships* between objects in the *real world*.

The most common types of functions (in MA140) are:

- *Linear* Functions, $f(x) = ax + b$
- *Polynomial* Functions, $x^3 + 3x^2 - 1$
- *Power* Functions, $f(x) = 3^x$
- *Rational* Functions, $f(x) = \frac{1+x}{1+x^2}$
- *Algebraic* Functions, $f(x) = \sqrt{x}$ or $f(x) = x^{3/2}$
- *Trigonometric* Functions, $f(x) = \sin(x)$
- *Exponential* Functions, $f(x) = e^x$.
- *Logarithms*. $f(x) = \log(x)$.

Linear functions have formulae such as $f(x) = mx + c$, where m and c are some given numbers.

It is often represented graphically as a straight line of slope m through the point $(0, c)$.



$f(x) = 2x - 1$

$f(0) = -1$

$f(1) = 1$

Polynomials

A **polynomial function** (or just **polynomial**) is a function of the form

$$y = f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x^1 + a_0, \quad x \in \mathbb{R},$$

where $a_0, a_1, \dots, a_n$ are real numbers called the **coefficients** of the polynomial.
The number n is called the **degree** of the polynomial.

There are special names for polynomials of low degree:

$n=0$ $\quad f(x) = a_0$ $\quad$ constant

$n=1$ $\quad f(x) = a_1 x + a_0$ $\quad$ linear

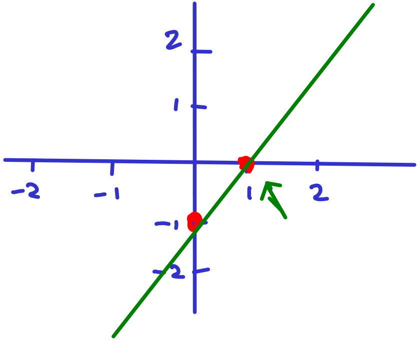
$n=2$ $\quad f(x) = a_2 x^2 + a_1 x + a_0$ $\quad$ quadratic

$n=3$ $\quad f(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$ $\quad$ cubic

Example: Linear Polynomial

A polynomial of degree $n = 1$ is called "linear". Its graph is a straight line. E.g. $y = x - 1$ is a **linear** polynomial.

$y(0) = -1$ $y(1) = 0$



FInished here Wednesday.

Example: quadratic

$x^2 - 2x - 3$ is a **quadratic** polynomial: it has degree $n = 2$.

There are many occasions when we want to **factorise** such quadratics, meaning we write them as the product of a pair of linear polynomials.

For example, we can **factorise** $x^2 - 2x - 3$ as

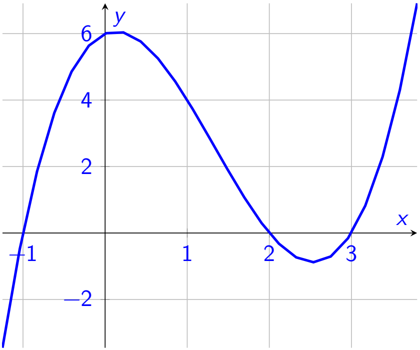
$$x^2 - 2x + 3 = (x - 3)(x + 1)$$

It is important to note that not all quadratic polynomials can be factorised as two linear polynomials. Such quadratics are called **irreducible**.

For example, $x^2 + 1$ is irreducible.

Example

$y = x^3 - 4x^2 + x + 6$ is a **cubic** function with degree $n = 3$.



Fact

A polynomial function of degree n has **up to** $n-1$ turning points ("bends").

Examples:

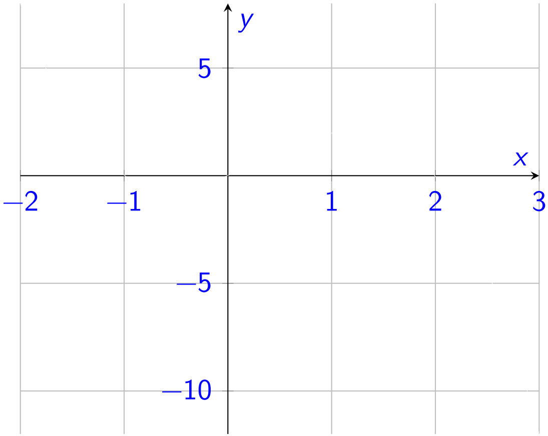
When sketching the graph of a function, we first find the **intercepts**:

- The y-**intercept** is where the graph of the function cuts the y-axis: found by letting $x = 0$.
- The x-**intercepts** are where the function's graph cuts the x-axis. These points are also called the **roots** (or **zeros**). To find them, set y equal to zero and solve for x.

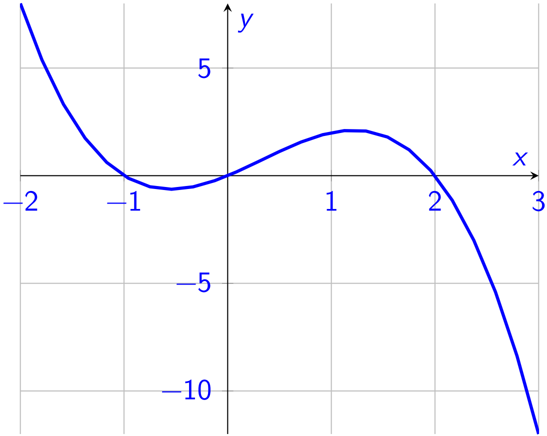
Example

Sketch the graph of $y = -x^3 + x^2 + 2x$

Now sketch $y = -x^3 + x^2 + 2x$



Actual plot of $y = -x^3 + x^2 + 2x$



Rational Functions

Rational Functions

Rational Functions have the general form

$$f(x) = \frac{p(x)}{q(x)},$$

where $p(x)$ and $q(x)$ are polynomials.

- If degree of $p(x) <$ degree of $q(x)$,
  $f(x)$ is called a **strictly proper rational function**.
- If degree of $p(x) =$ degree of $q(x)$,
  $f(x)$ is called a **proper rational function**.
- If degree of $p(x) >$ degree of $q(x)$,
  $f(x)$ is called an **improper rational function**.

Rational Functions

An improper or proper rational function can always be expressed as a polynomial plus a strictly proper rational function, for example by algebraic division.

Example

$$\frac{4x^3+4x^2+4}{x^2-3} = 4x+4+\frac{12x+16}{x^2-3}$$

For the previous example, we can work this out ourselves using **Long Division** to divide numerator by denominator:

Example 2.30 from text book

Use long division to show that

$$\frac{3x^4+2x^3-5x^2+6x-7}{x^2-2x+3} = 3x^2+8x+2-\frac{14x+13}{x^2-2x+3}$$

Exercises

Exercise 1.2.1

Identify the largest possible subset of $\mathbb{R}$ that could be the domain and range of these functions:

1. $f(x) = (x-4)^2 + 5$
2. $f(x) = \sqrt{3x+2} - 1$
3. $f(x) = 3/(x-2)$.

(See Example 1.1.2 of the textbook).

Exercise 1.2.2

Sketch the graphs of

(i) $y = 5x^2 - 7$

(ii) $y = x^2 - 4x + 3$

(iii) $y = x^3 - 6x^2 - 11x - 6$